

ACTIVITY

TOOL: R Programming

RegisterNo:192424189

Name: Dhanusha D

Date:24-07-2026

COURSE: Data Handling and Visualization

Code: DSA0604

DATASET: Employee Salary Dataset

TYPES OF VISUALIZATION

1.AMOUNTS:

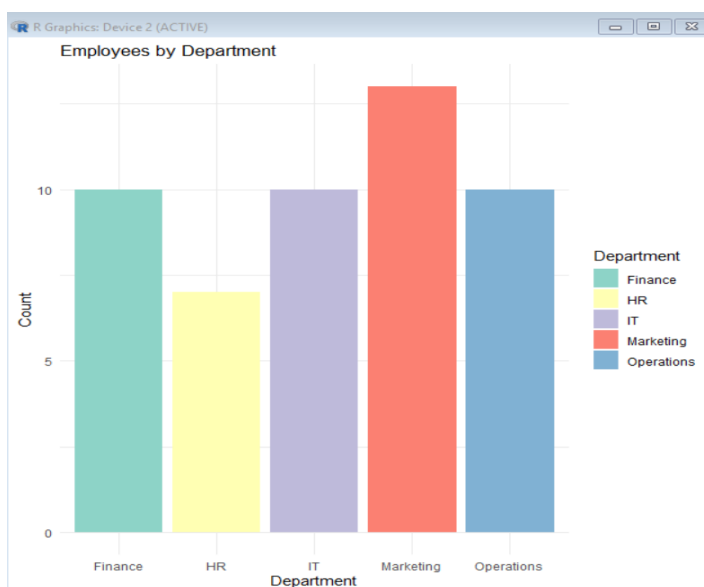
A. BAR PLOT

CODE:

```
library(ggplot2)

ggplot(data, aes(x = Department, fill = Department)) +
  geom_bar() +
  scale_fill_brewer(palette = "Set3") +
  labs(title = "Employees by Department",
       x = "Department",
       y = "Count") +
  theme_minimal()
```

OUTPUT:



B. DOT PLOT

CODE:

```
ggplot(data,  
  aes(x = Monthly_Salary,  
    fill = Department)) +  
geom_dotplot(binwidth = 5000,  
  method = "histodot",  
  stackdir = "center") +  
scale_fill_brewer(palette = "Dark2") +  
labs(title = "Dot Plot of Monthly Salary",  
  x = "Monthly Salary",  
  y = "Count") +  
theme_minimal()
```

OUTPUT:

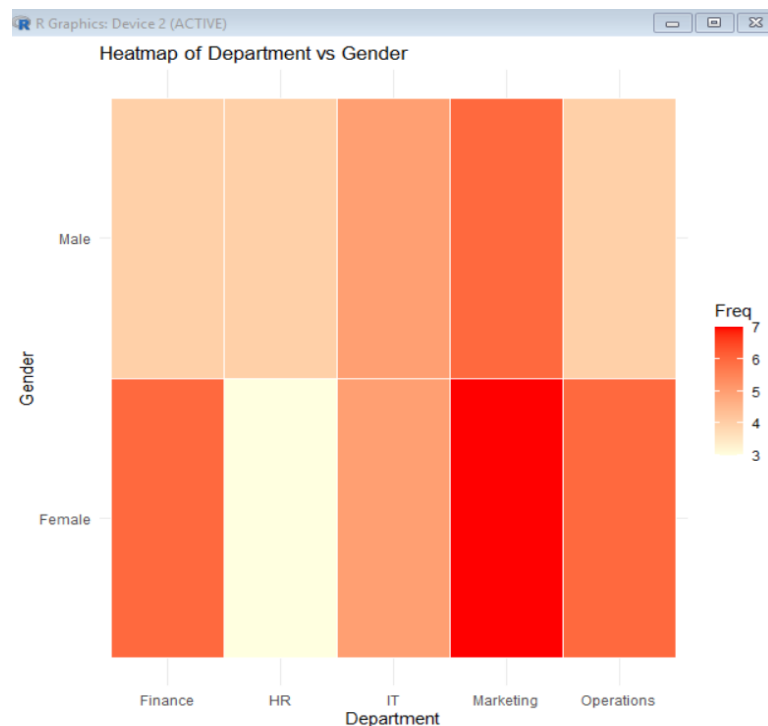


C. HEATMAP

CODE:

```
heat <- as.data.frame(table(data$Department,  
                             data$Gender))  
  
ggplot(heat,  
       aes(x = Var1,  
           y = Var2,  
           fill = Freq)) +  
  geom_tile(color = "white") +  
  scale_fill_gradient(low = "lightyellow",  
                     high = "red") +  
  labs(title = "Heatmap of Department vs Gender",  
       x = "Department",  
       y = "Gender") +  
  theme_minimal()
```

OUTPUT:

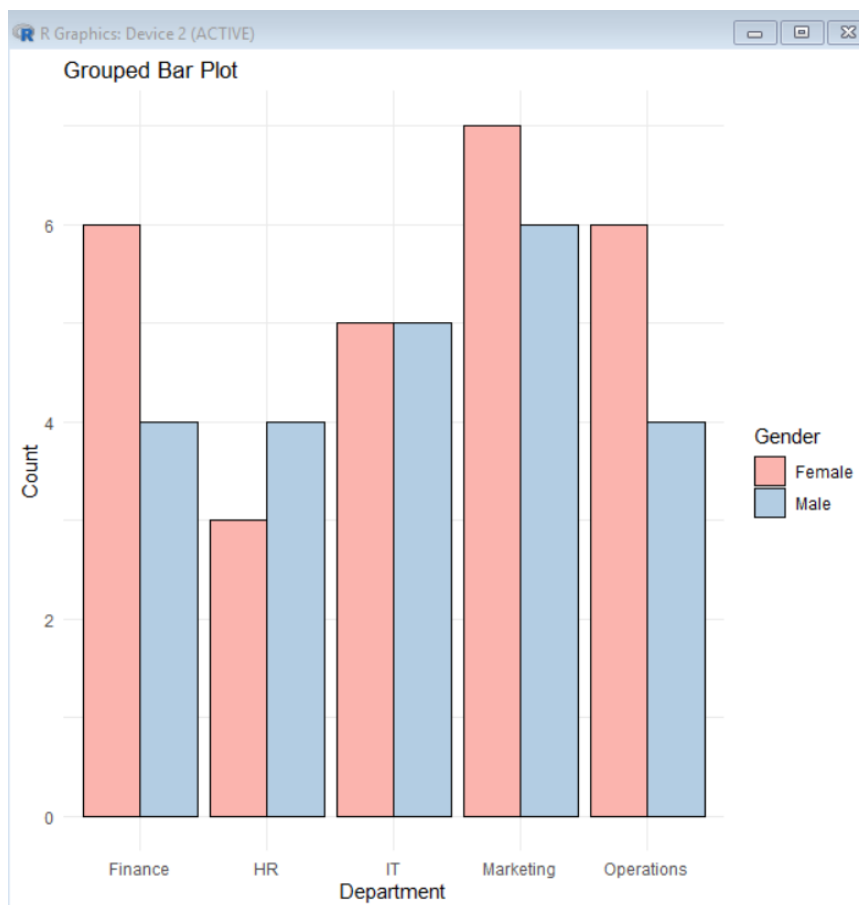


D. GROUPED BAR PLOT

CODE:

```
ggplot(data,  
  aes(x = Department,  
    fill = Gender)) +  
geom_bar(position = "dodge",  
  color = "black") +  
scale_fill_brewer(palette = "Pastel1") +  
labs(title = "Grouped Bar Plot",  
  x = "Department",  
  y = "Count") +  
theme_minimal()
```

OUTPUT:



E. STACKED BAR PLOT

CODE:

```
ggplot(data,  
  aes(x = Department,  
    fill = Gender)) +  
  geom_bar(color = "black") +  
  scale_fill_brewer(palette = "Set2") +  
  labs(title = "Stacked Bar Plot",  
    x = "Department",  
    y = "Count") +  
  theme_minimal()
```

OUTPUT:



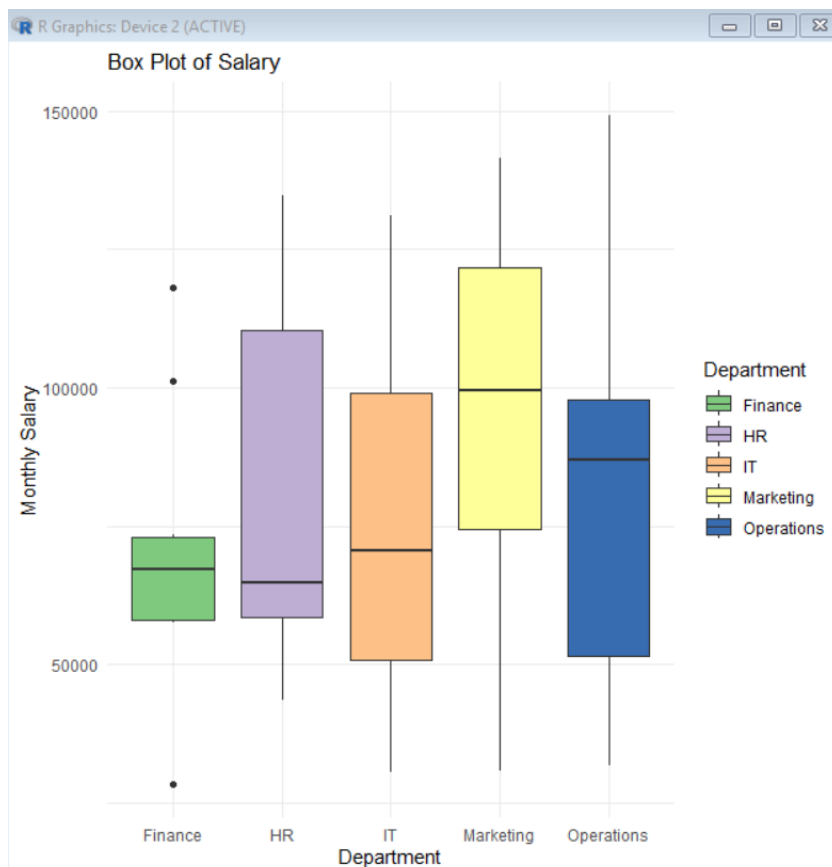
2.DISTRIBUTIONS

A. BOX PLOT

CODE:

```
ggplot(data,  
  aes(x = Department,  
    y = Monthly_Salary,  
    fill = Department)) +  
geom_boxplot() +  
scale_fill_brewer(palette = "Accent") +  
labs(title = "Box Plot of Salary",  
  x = "Department",  
  y = "Monthly Salary") +  
theme_minimal()
```

OUTPUT:

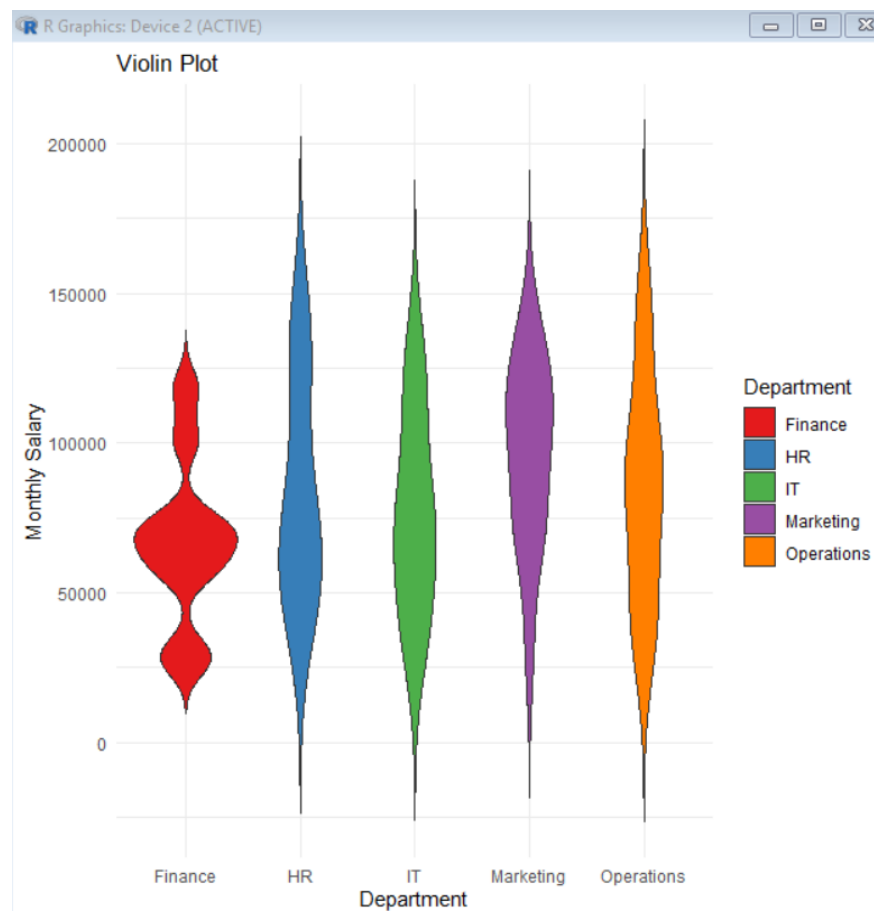


B. VIOLIN PLOT

CODE:

```
ggplot(data,  
  aes(x = Department,  
    y = Monthly_Salary,  
    fill = Department)) +  
geom_violin(trim = FALSE) +  
scale_fill_brewer(palette = "Set1") +  
labs(title = "Violin Plot",  
  x = "Department",  
  y = "Monthly Salary") +  
theme_minimal()
```

OUTPUT:



C. HISTOGRAM

CODE:

```
library(RColorBrewer)

ggplot(data, aes(x = Monthly_Salary, fill = after_stat(count))) +
  geom_histogram(binwidth = 5000, color = "black") +
  scale_fill_distiller(palette = "Spectral") +
  labs(title = "Histogram of Monthly Salary") +
  theme_minimal()
```

OUTPUT:

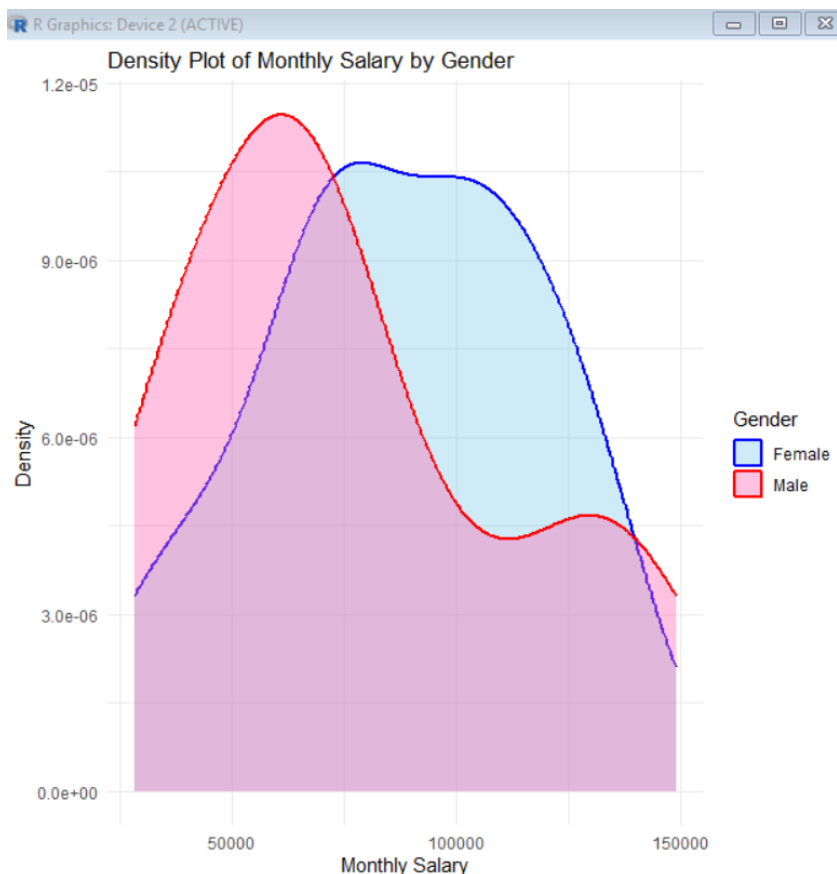


D. DENSITY PLOT

CODE:

```
ggplot(data,  
  aes(x = Monthly_Salary,  
    fill = Gender,  
    color = Gender)) +  
  geom_density(alpha = 0.4, size = 1) +  
  scale_fill_manual(values = c("skyblue", "hotpink")) +  
  scale_color_manual(values = c("blue", "red")) +  
  labs(title = "Density Plot of Monthly Salary by Gender",  
    x = "Monthly Salary",  
    y = "Density") +  
  theme_minimal()
```

OUTPUT:

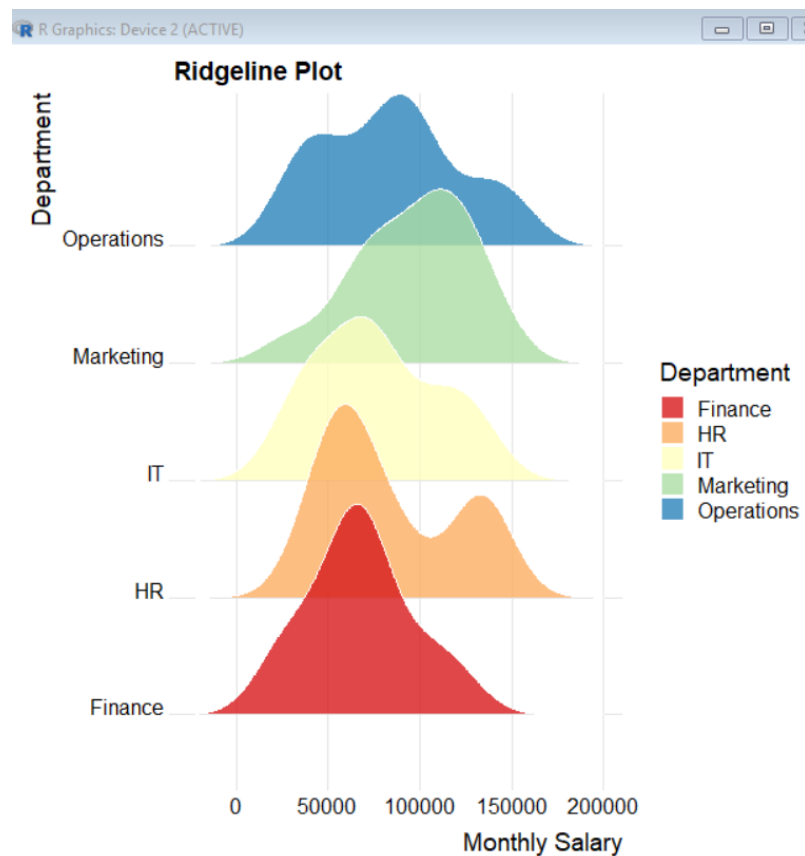


E. RIDGELINE /(JOY) PLOT

CODE:

```
ggplot(data,  
  aes(x = Monthly_Salary,  
    y = Department,  
    fill = Department)) +  
  geom_density_ridges(alpha = 0.8,  
    color = "white") +  
  scale_fill_brewer(palette = "Spectral") +  
  labs(title = "Ridgeline Plot",  
    x = "Monthly Salary",  
    y = "Department") +  
  theme_ridges()
```

OUTPUT:



3.PROPORTIONS

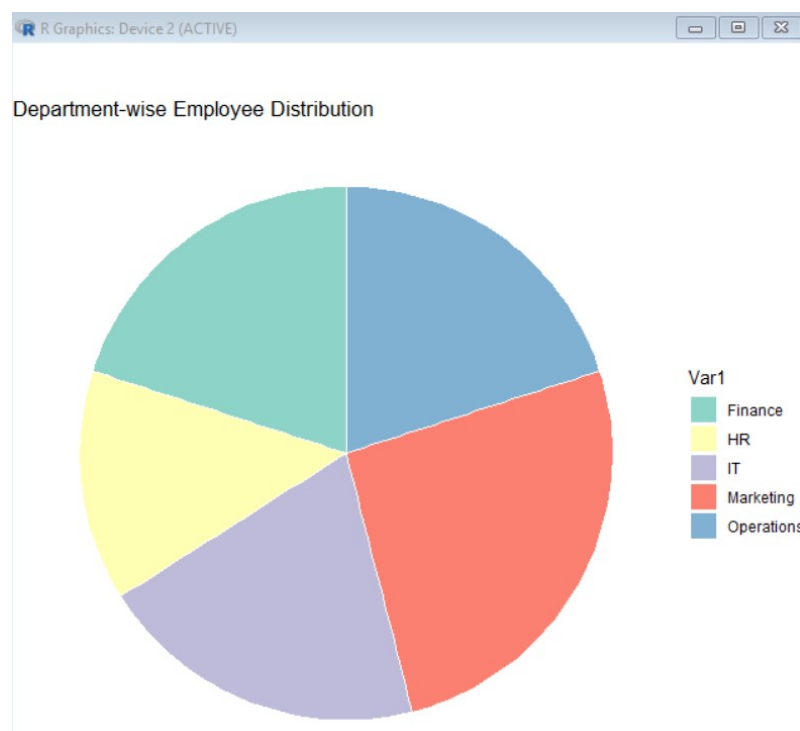
A. PIE CHART

CODE:

```
dept <- as.data.frame(table(data$Department))

ggplot(dept,
       aes(x = "",
           y = Freq,
           fill = Var1)) +
  geom_bar(stat = "identity",
           width = 1,
           color = "white") +
  coord_polar("y") +
  scale_fill_brewer(palette = "Set3") +
  labs(title = "Department-wise Employee Distribution") +
  theme_void()
```

OUTPUT:



B. STACKED BARS

CODE:

```
library(ggplot2)

ggplot(data,

  aes(x = Department,

      fill = Education_Level)) +

  geom_bar(color = "black") +

  scale_fill_brewer(palette = "Set3") +

  labs(title = "Stacked Bar Chart of Department by Education Level",

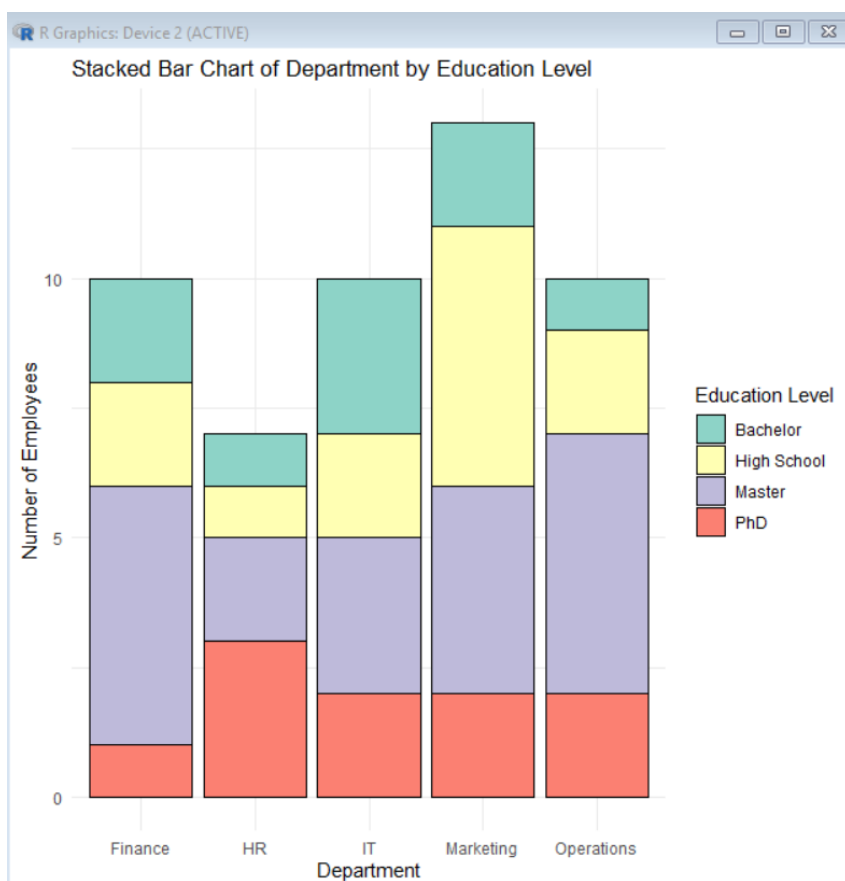
       x = "Department",

       y = "Number of Employees",

       fill = "Education Level") +

  theme_minimal()
```

OUTPUT:



C. TREEMAP

CODE:

```
dept <- as.data.frame(table(data$Department))  
treemap(  
  dept,  
  index = "Var1",  
  vSize = "Freq",  
  type = "index",  
  palette = "Set3",  
  title = "Department Treemap"  
)
```

OUTPUT:

